

DIGITAL QUEUE MANAGEMENT SYSTEM (DQMS)

Enterprise Platform User Guide & Operations Manual (Stages 1 - 3)

Product Version: v1.0.0 Enterprise	Technology Stack: .NET 8 Web API + Dapper ORM
Frontend Framework: Flutter (Riverpod/Dio)	Databases: MySQL 8.0 & MS SQL Server 2022
Design Standard: Command Center Spec	Architecture: Service Pattern (No Repositories)

1. Executive System Overview

The **AntiGravity Enterprise DQMS** is a multi-tenant, high-performance Queue Management System engineered for operational efficiency, zero-friction operator workflows, and real-time waiting room displays. Built using a **C# .NET Core Web API backend** with **Dapper ORM stored procedures** and a cross-platform **Flutter frontend**, the system operates across 3 distinct perspectives:

Stage / Perspective	Target User Group	Core Capability & Features
Stage 1: Admin Setup	Application Administrators	Master configuration for Areas/Zones, Process Pipelines with SLA TAT, Counter Windows, and TV Display Templates.
Stage 2: Staff Console	Counter Operators / Staff	High-speed, low-click operator station powered by direct keyboard hotkeys (Space / F1 to F7) and priority queueing.
Stage 3: Customer Module	Patients & Customers	4K Overhead TV Display Boards, Touchscreen Ticket Kiosk, Mobile Web QR Ticket Tracking, and WhatsApp alerts.

2. Stage 1: Application Admin Masters

Stage 1 provides administrators with structured data tables and real-time KPI metrics for configuring organizational queue infrastructure. Access the panel by selecting **Admin Master Configuration**.

Key Admin Master Views:

Master Entity	Configured Parameters	Database Stored Procedures
Areas & Zones	Area Code, Area Name, Location ID, Active Status	PR_S_Area, PR_IU_Area
Process Pipelines	Token Prefix (A-Z), Target SLA TAT (mins), Sub-Tokens	PR_S_Process, PR_IU_Process

Counter Stations	Counter Number, Window Name, Status Code (20001)	
		PR_S_Counter, PR_IU_Counter
Display Templates	Layout Type (GridView 21001, Split 21002, List 21003)	
		PR_S_DisplayTemplate, PR_IU_DisplayTemplate

3. Stage 2: Staff Counter Station & Keyboard Hotkeys

The **Counter Operator Console** is engineered for high-speed operation. Staff members can call, recall, serve, hold, complete, and cancel queue tickets without using the mouse by leveraging physical keyboard hotkeys.

Operator Keyboard Hotkey Quick Reference:

Hotkey	Action Name	Target Token Status	Description & Behavior
SPACE / F1	Call Next	Calling (18003)	Automatically pulls highest priority waiting token (VIP, Senior, Standard).
F2	Recall	Calling (18003)	Triggers audio announcement & TV visual flash pulse again.
F3	Serve Active	Active (18004)	Marks customer active at counter and starts SLA service timer.
F4	Hold	Hold (18005)	Places ticket on hold with optional operator reason notes.
F5	Complete & Call Next	Completed (18007)	Completes active ticket and immediately calls next waiting customer.
F7	Cancel	Canceled (18006)	Cancels ticket for no-show or customer departure.

4. Stage 3: Customer Displays, Kiosk & Mobile Tracking

Stage 3 includes 3 dedicated public interfaces designed for customer engagement and waiting room communication:

Interface View	Access Point	Key Features
4K Overhead TV Display Board	Overhead TV Monitors (Web Browser)	GridView layout (Template 21001) displaying NOW CALLING tokens (e.g. A-001 -> COUNTER 03) with 30-second green/red visual pulse alerts and audio announcements.
Self-Service Ticket Kiosk	Touchscreen Kiosk Terminal	Large touch buttons for service selection, category filter (Standard, Senior Citizen, Disabled, VIP), and thermal ticket printer integration.
Mobile Web Ticket Tracker	Customer Smartphone (QR Code Scan)	Live position in queue tracker showing exact Customers Ahead (e.g. 2 ahead) and Estimated Wait Time (~15 mins) with auto-refresh.

5. API Endpoints & SQL Stored Procedures Reference

HTTP Method & Endpoint	Controller Action	Stored Procedure
GET /api/v1/admin/areas	Get Areas by Org & Loc	PR_S_Area
POST /api/v1/admin/area	Create / Update Area	PR_IU_Area

POST /api/v1/staff/issue-token	Issue Queue Ticket	PR_IU_IssueToken
POST /api/v1/staff/call-next	Operator Call Next Token	PR_IU_CallNextToken
POST /api/v1/staff/update-status	Update Token Status	PR_IU_UpdateTokenStatus
GET /api/v1/public/display-board	Overhead TV Board Stream	PR_S_PublicDisplayBoard
GET /api/v1/public/ticket-status/{id}	Mobile QR Ticket Status	PR_S_PublicTokenStatus

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